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Triglyceride glucose-weight-adjusted waist index as a cardiovascular mortality predictor: incremental value beyond the establishment of TyG-related indices

Jiajun Qiu^{1,2}, Jin'e Li^{1,2}, Shan Xu^{1,2}, Jingqi Yang³, Haixia Zeng^{1,2,4}, Yuying Zhang^{1,2}, Shiqi Yang^{1,2}, Lixuan Fang^{1,2}, Jiadian Huang^{1,2}, Hongtao Zhou^{1,2}, Jiaying Feng^{1,2}, Yujie Zan^{1,2}, Jia Zhan^{1,2} and Jianping Liu^{1,2,4,5*}

Abstract

Background Cardiovascular disease (CVD) remains the leading cause of mortality worldwide, emphasizing the need for enhanced risk stratification tools. The triglyceride–glucose-weight adjusted waist index (TyG-WWI), which integrates insulin resistance and central obesity, has emerged as a potential predictor, but its performance relative to traditional TyG-related indices (TyG, TyG-WC, TyG-WHtR) requires further validation.

Methods We analyzed data from 24,255 participants in the National Health and Nutrition Examination Survey (NHANES, 1999–2018). Weighted Cox proportional hazards models were used to assess the associations between TyG-related indices and cardiovascular mortality. Restricted cubic splines (RCSs) with four knots were employed to explore dose–response relationships. Traditional and time-dependent receiver operating characteristic (ROC) analyses, net reclassification improvement (NRI) analyses, and subgroup and sensitivity analyses were conducted to evaluate predictive performance and robustness.

Results Over a median follow-up of 9.67 years, 854 cardiovascular deaths were recorded. According to the fully adjusted models, each increase in the standard deviation of the TyG-WWI was associated with a 45% greater risk of cardiovascular mortality (HR = 1.45, 95% CI 1.31–1.60), which was stronger than the associations observed for TyG (HR = 1.24, 95% CI 1.12–1.38), TyG-WC (HR = 1.39, 95% CI 1.27–1.53), and TyG-WHtR (HR = 1.43, 95% CI 1.30–1.58). When stratified by quartiles, the TyG-WWI exhibited a clear dose–response relationship. RCS analyses revealed that the TyG-WWI had a linear association with cardiovascular mortality ($P_{\text{nonlinear}} = 0.491$), whereas the TyG index exhibited a U-shaped association, and the TyG-WC index and TyG-WHtR showed L-shaped associations (all $P_{\text{nonlinear}} < 0.05$). Traditional ROC analysis revealed that the TyG-WWI had the highest AUC (0.694, 95% CI 0.678–0.710). Time-dependent ROC analyses demonstrated that the AUC for the TyG-WWI ranged from 0.706 to 0.751 across different follow-up time points, which was consistently greater than those of the other TyG-related indices. NRI analyses indicated significant improvements in risk reclassification when the TyG-WWI was used compared with traditional TyG-related indices.

*Correspondence:
Jianping Liu
ndefy14105@ncu.edu.cn

Full list of author information is available at the end of the article



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(10.4% vs. TyG, 9.4% vs. TyG-WC, 9.1% vs. TyG-WHtR). Subgroup analyses revealed stronger associations in younger adults (≤ 60 years, HR = 2.03, 95% CI 1.78–2.32).

Conclusion The current study is the first to validate that the TyG-WWI is a reliable risk prediction tool for cardiovascular death in the general population and has greater predictive value than traditional TyG-related parameters. The results support its potential as a supplementary tool among TyG-derived markers for assessing cardiovascular death.

Keywords Cardiovascular mortality, Insulin resistance, Incremental value, TyG index, TyG-WC, TyG-WHtR, TyG-WWI

Introduction

Cardiovascular disease (CVD) remains a leading cause of mortality worldwide, with increasing incidence rates and mortality rates imposing a substantial public health burden [1–3]. According to the latest global burden of disease data, the number of CVD cases increased by 341 million from 1990 to 2021 (rising from 271 to 612 million), accompanied by an additional 7.3 million deaths (with total deaths increasing from 12.1 to 19.4 million) [1, 4]. Consequently, the effective identification of high-risk populations for cardiovascular mortality is critical for implementing targeted prevention strategies [5]. While traditional risk factors such as hypertension, diabetes, and dyslipidemia are well-established predictors of CVD [4], they often fail to fully capture the complex interplay between metabolic and anthropometric parameters underlying CVD risk.

Insulin resistance (IR) has emerged as a pivotal factor in the pathophysiology of CVD [6, 7], highlighting the need for simple and reliable IR indicators to enhance risk stratification. The triglyceride–glucose (TyG) index, which is derived from fasting triglyceride and glucose levels, has been proposed as a surrogate marker of IR and is associated with an elevated risk of incident CVD and mortality across diverse populations [8–14]. However, the TyG index does not account for obesity, a key determinant of both IR and incident CVD [15, 16]. To address this limitation, several TyG-related obesity indices (e.g., TyG-WC and TyG-WHtR) have been developed and investigated as potential predictors of incident CVD risk [17–27].

Nevertheless, controversy remains regarding whether combining TyG with central obesity indices yields superior predictive performance for cardiovascular mortality compared with the TyG index alone—particularly in populations with metabolic diseases [24, 28–31]. One potential explanation is that waist circumference (WC) and the waist-to-height ratio (WHtR) fail to comprehensively reflect central obesity in individuals with preexisting metabolic abnormalities. Additionally, given the obesity paradox, single anthropometric indices cannot be used to assess fat distribution fully [32–35]. The weight-adjusted waist index (WWI), a novel marker of central obesity, integrates waist circumference (reflecting abdominal fat) and weight (reflecting overall adiposity), thereby

optimizing fat distribution assessment and potentially improving incident CVD risk evaluation in clinical settings [36]. Prior studies have demonstrated that the WWI is correlated with all-cause and cardiovascular mortality in diverse U.S. and Chinese populations [37–42] and outperforms traditional anthropometric indices (e.g., BMI and WC) in predicting incident CVD [43]. The recently proposed TyG-WWI further optimizes the combination of these factors [44]. By integrating the TyG index with the WWI (which reflects central fat distribution), the TyG-WWI may more comprehensively assess an individual's metabolic status and fat distribution characteristics, potentially exhibiting superior predictive potential for CVD risk. To date, only one study has evaluated the association between the TyG-WWI and mortality in individuals with metabolic dysfunction-associated steatotic liver disease (MASLD) and prediabetes or diabetes [28]. However, no studies have examined the relationship between the TyG-WWI and cardiovascular mortality in the general population. Additionally, comparative analyses of the predictive value of the TyG-WWI versus other TyG-related obesity indices for cardiovascular mortality remain limited, with few studies directly assessing their predictive accuracy.

This study aimed to investigate the association between the TyG-WWI and cardiovascular mortality in the general population and to compare its predictive accuracy with that of conventional TyG-related parameters in a cohort of U.S. adults from the National Health and Nutrition Examination Survey (NHANES). We hypothesized that the TyG-WWI would demonstrate superior predictive performance for cardiovascular mortality compared with traditional TyG-related indices. These findings may provide valuable insights for refining risk stratification and developing targeted strategies for CVD prevention.

Methods

Data sources and study population

The data for this study were derived from the National Health and Nutrition Examination Survey (NHANES), encompassing 10 survey cycles conducted between 1999 and 2018. Detailed information about the NHANES is available on its official website [45]. Originally designed as a cross-sectional survey to assess the health and

nutritional status of the U.S. population, the NHANES employs a multistage probability sampling design to ensure national representativeness. Through linkage with the National Death Index (NDI), the dataset was transformed into a prospective cohort dataset suitable for longitudinal analyses.

In our study, the cohort follow-up period commenced on the date of each participant's initial NHANES examination (baseline) and concluded at the earliest of three time points: (1) the date of death due to cardiovascular disease (CVD), defined by ICD-10 codes I00–I99 as recorded in NDI death certificates; (2) the date of death from non-CVD causes; or (3) December 31, 2019 (the end of the follow-up period). Time-to-event data were processed by calculating each participant's follow-up duration as the interval from their baseline examination to their respective endpoint (CVD death) or censoring event (nCVD death, loss to follow-up, or end of follow-up).

A total of 101,316 participants from the 1999 to 2018 cycles were initially identified. To ensure data

completeness and validity, the following participants were excluded: (1) 42,112 participants aged <18 years; (2) 140 participants with missing mortality data (corresponding to a loss-to-follow-up rate of approximately 0.24% (140/59,204)), including individuals whose vital status or cause of death could not be confirmed via NDI linkage by the end of follow-up (December 31, 2019) (distinct from participants confirmed to be alive as of the cutoff date, who were retained and censored); (3) 33,724 participants with missing TyG index data; and (4) 1,085 participants with missing measurements of height, weight, or waist circumference. Ultimately, 24,255 participants were included in the final analysis. A detailed flowchart of participant inclusion is provided in Fig. 1.

Definition of terminal events

The primary endpoint of this study was CVD-related mortality. Mortality data, including information on cardiovascular deaths, were obtained via linkage with the National Death Index (NDI), a system overseen and maintained by the National Center for Health Statistics

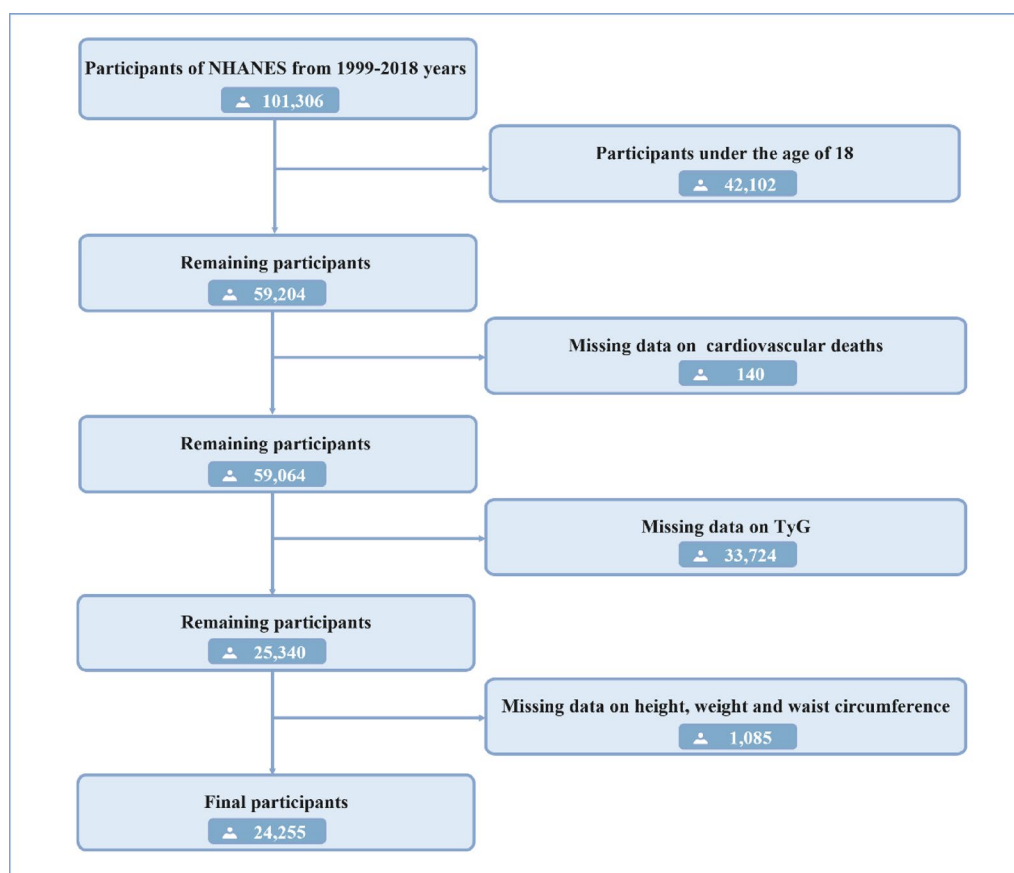


Fig. 1 Study population screening flowchart. The NHANES data were screened on the basis of the study design to select eligible participants. Note: In our study, "missing data on cardiovascular deaths" specifically refer to participants whose vital status (alive or deceased) or cause of death could not be confirmed via the NDI linkage by the end of our follow-up period (December 31, 2019). This includes two scenarios: (1) participants for whom NDI records were unavailable due to incomplete linkage (e.g., mismatched identifiers) and (2) individuals who were lost to follow-up before mortality status could be verified

(NCHS). This linkage utilizes multiple identifiers (e.g., social security number, date of birth) to optimize tracking accuracy, thereby ensuring high reliability; it is widely recognized as the gold standard for mortality ascertainment in large-scale epidemiological studies [46]. For the specific classification of cardiovascular deaths, the International Classification of Diseases, Tenth Revision (ICD-10) clinical modification codes were applied. After death records from the NDI were retrieved, trained coders systematically reviewed each record to identify cases where the underlying cause of death was classified under CVD-related ICD-10 codes. These included codes for heart diseases (I00–I09, I11, I13, I20–I51) and cerebrovascular diseases (I60–I69), encompassing a comprehensive range of conditions such as ischemic heart disease, cerebrovascular disease, and other disorders of the heart and circulatory system [13, 47]. The mortality follow-up period extended through December 31, 2019. The observation period for each participant was defined as the interval from their initial NHANES enrollment to either the occurrence of a CVD-related death or the study cut-off date, whichever came first.

Definitions of TyG, TyG-WC, TyG-WHtR, and TyG-WWI

TyG, TyG-WC, TyG-WHtR, and TyG-WWI are calculated via the following formulas [8, 17, 44]:

$$\text{TyG} = \ln (\text{fasting triglyceride (mg/dL)}) \times \text{fasting glucose (mg/dL)} / 2$$

$$\text{TyG} - \text{WC} = \text{TyG} \times \text{Waist circumference (cm)}.$$

$$\text{TyG} - \text{WHtR} = \text{TyG} \times (\text{waist circumference (cm)} / \text{height (cm)})$$

$$\text{TyG} - \text{WWI} = \text{TyG} \times \frac{\text{Waist Circumference (cm)}}{\sqrt{\text{Weight (kg)}}}.$$

Fasting triglyceride and glucose levels were obtained from blood samples collected at mobile examination centers after an 8-h fasting period. Body weight, height, and waist circumference were measured by trained professionals at these centers.

Covariates

To ensure the accuracy and reliability of the analytical results, this study comprehensively incorporated multiple potential covariates associated with cardiovascular disease mortality risk. These covariates were selected on the basis of well-established cardiovascular risk factors and prior literature, including demographic characteristics, lifestyle factors, and medical history, as detailed below: Demographic characteristics included age (continuous

variable), sex (male/female), race/ethnicity (Hispanic White, non-Hispanic Black, Mexican, and other ethnicities), educational attainment (less than high school or equivalent, college or above), marital status (married/living with a partner, widowed/divorced/separated, never married), and poverty-income ratio (PIR), which was categorized as low ($\text{PIR} < 1.35$), medium ($1.35 \leq \text{PIR} < 3.0$), and high ($\text{PIR} \geq 3.0$). Lifestyle-related factors encompassed alcohol consumption status (never, former, current; current drinkers were further stratified into mild, moderate, and heavy) and smoking status (never, former, current). Medical history was systematically recorded, including coronary heart disease (CHD), determined by self-reported responses to the questionnaire item “Has a doctor or other health professional ever told you that you had coronary heart disease?”; hypertension, defined by physician diagnosis, documented use of antihypertensive drugs, or two separate blood pressure measurements exceeding diagnostic thresholds; and stroke history, assessed via questionnaire responses to “Has a doctor or other health professional ever told you that you had a stroke?” Diabetes mellitus (DM): This condition is characterized by self-reported diagnosis, self-reported use of medications or insulin to lower blood glucose, an HbA1c level of 6.5% or higher, or a fasting plasma glucose level of ≥ 7.0 mmol/L. Hyperlipidemia is defined as triglycerides of 150 mg/dl or higher, total cholesterol of 200 mg/dl or higher, LDL of 130 mg/dl or higher without the use of cholesterol-lowering drugs, HDL lower than 40 mg/dl in men, or HDL lower than 50 mg/dl in women [48].

To address missing data while preserving the integrity of the observed data, a dual analytical strategy was employed. Given that all covariates with missing values were categorical (e.g., educational attainment, smoking status), the primary analysis adopted a complete-case approach combined with enhanced categorical reclassification: missing values for each categorical variable were redefined as a new “unknown” subgroup. This approach maximized the utilization of available observed data and avoided artificial patterns that might be introduced by imputation. Detailed information on covariate missingness is presented in Supplementary Table 1. Additionally, to mitigate potential bias arising from missing data, comprehensive sensitivity analyses were conducted in accordance with statistical standards, using multiple imputation (MI) to generate 4 imputed datasets.

Statistical analysis

Given the complex multistage sampling design of the NHANES data, all analyses were performed via the weighting methodology recommended by the NHANES. Since this study incorporated blood biochemical data (including glucose and lipid indicators required for TyG index calculation) and involved data synthesis across

multiple survey cycles, mobile examination center (MEC) examination weights were applied in accordance with NHANES analytical guidelines. Specifically, the Full Sample 4-Year MEC Exam Weight (WTMEC4YR) was used for the two survey cycles from 1999 to 2002, whereas the Full Sample 2-Year MEC Exam Weight (WTMEC2YR) was adopted for the eight survey cycles from 2003 to 2018. This weighting strategy ensures that the findings are representative of the noninstitutionalized civilian population of the United States. These weights were explicitly incorporated into key analyses, including Cox regression and competing risk models, to account for the complex sampling design. For receiver operating characteristic (ROC) analysis and restricted cubic spline (RCS) analysis, weights were not applied, as there is no consensus in the statistical community on standardized weighting procedures for such analyses within complex sampling frameworks.

First, group comparisons were performed on the basis of cardiovascular mortality status. After weighted analyses, all continuous variables exhibited a normal distribution; thus, continuous variables are expressed as the mean (standard) deviation, and categorical variables are expressed as the frequency (%), which is consistent with previously published methodologies [49]. Weighted linear regression was used to compare continuous variables, whereas weighted chi-square tests were applied for categorical variables. To comprehensively evaluate the potential impact of baseline data on the results, several supplementary analyses were conducted. First, considering the potential influence of the distribution characteristics of TyG-related parameters on the findings, their distribution patterns were assessed via multiple normality tests (Anderson–Darling test, Cramer–von Mises test, Lilliefors test, and Pearson’s chi-square test) combined with visual inspections (histograms and Q–Q plots). Additionally, to address potential selection bias introduced by participants excluded because of missing data, baseline characteristics were compared between participants with complete data and those excluded because of missing data. A standardized difference exceeding 10% was considered indicative of a significant discrepancy [50, 51].

To evaluate the associations between TyG-related indices (TyG, TyG-WC, TyG-WHtR, and TyG-WWI) and the risk of cardiovascular mortality, three multivariate Cox proportional hazards regression models were constructed: a crude model without covariate adjustment; Model I adjusted solely for age and sex; and Model II, which incorporated all covariates from Model I plus ethnicity, education, marital status, PIR, drinking status, smoking status, and histories of CHD, DM, hyperlipidemia, and hypertension. Prior to finalizing Model II, collinearity diagnostics using variance inflation factors

(VIFs) were performed (Supplementary Tables 2–5), leading to the exclusion of stroke history due to significant collinearity with CHD ($VIF > 5$). All four indices were standardized via Z score transformation prior to Cox regression analysis to quantify the cardiovascular mortality risk associated with each standard deviation increment. Additionally, in all three models, the indices were stratified by quartiles, and trend tests were conducted to assess changes in cardiovascular mortality risk across increasing levels of the indices. The proportional hazards assumption was verified via the Schoenfeld residual method (Supplementary Fig. 1) [52], with p values of 0.531, 0.056, 0.225, and 0.218 for the models evaluating TyG, TyG-WC, TyG-WHtR, and TyG-WWI, respectively. All p values exceeded 0.05, confirming that the proportional hazards assumption was satisfied for all the models.

To explore potential nonlinear relationships, we employed an RCS with four knots (placed at the 5th, 35th, 65th, and 95th percentiles) to model the dose–response associations between TyG, TyG-WC, TyG-WHtR, TyG-WWI, and cardiovascular mortality; these models were adjusted for the same covariates as those in Model II. The nonlinearity of associations was evaluated via likelihood ratio tests by comparing the RCS model with a simple linear model. For indices where nonlinear relationships were identified (P for nonlinearity < 0.05), we further performed two-piecewise linear regression to determine inflection points, which were calculated via a recursive algorithm. Separate hazard ratios (HRs) and 95% confidence intervals (CIs) were estimated for the segments above and below each inflection point, with adjustment for the same covariate set to ensure consistency with primary analyses (Table 1).

ROC curve analysis was performed to evaluate the predictive value of TyG, TyG-WC, TyG-WHtR, and TyG-WWI for cardiovascular mortality, with the area under the curve (AUC) and corresponding 95% CIs used to quantify the discriminative ability of each index. The DeLong test was applied to compare the AUC of the TyG-WWI with those of the other three TyG-related indices to determine whether the TyG-WWI has superior predictive performance. Given the potential inflation of type I error due to multiple comparisons (encompassing three pairwise tests), Bonferroni correction was implemented to adjust the significance threshold to $\alpha = 0.05/3 \approx 0.017$. Furthermore, net reclassification improvement (NRI) analysis was conducted to quantify the proportion of correct reclassifications in individual risk categories for cardiovascular death risk stratification when the TyG-WWI was used compared with traditional TyG-related indices, thereby assessing its incremental clinical value. As a complement to the predictive models, Hosmer–Lemeshow tests and calibration curves were generated;

Table 1 Baseline characteristics of participants stratified by the presence of cardiovascular death

Characteristic	Alive	Death	P value
Participants	23,401	854	
Age, years	45.61 (0.22)	67.62 (0.69)	< 0.01
Weight, kg	81.82 (0.22)	83.27 (0.98)	0.19
Height, cm	169.05 (0.09)	167.58 (0.47)	0.01
BMI, kg/m ²	28.55 (0.08)	29.49 (0.29)	< 0.01
WC, cm	97.84 (0.20)	104.22 (0.68)	< 0.01
TyG	8.61 (0.01)	8.94 (0.03)	< 0.01
TyG-WC	846.42 (2.22)	934.78 (7.21)	< 0.01
TyG-WHtR	5.01 (0.01)	5.58 (0.04)	< 0.01
TyG-WWI	93.75 (0.16)	102.92 (0.46)	< 0.01
Sex			< 0.01
Female	12,100 (51.71%)	345 (40.40%)	
Male	11,301 (48.29%)	509 (59.60%)	
Ethnicity			< 0.01
Non-Hispanic White	10,030 (42.86%)	519 (60.77%)	
Non-Hispanic Black	4777 (20.41%)	173 (20.26%)	
Mexican American	4454 (19.03%)	105 (12.30%)	
Other races	4140 (17.69%)	57 (6.67%)	
Education			0.32
Below high school	5616 (24.00%)	198 (23.19%)	
High school or equivalent	6802 (29.07%)	240 (28.10%)	
Above high school	7059 (30.17%)	252 (29.51%)	
Unknown	3924 (16.77%)	164 (19.20%)	
Marital status			0.29
Married/cohabiting	7547 (32.25%)	263 (30.80%)	
Widowed/divorced/separated	2887 (12.34%)	98 (11.48%)	
Never married	3663 (15.65%)	153 (17.92%)	
Unknown	9304 (39.76%)	340 (39.81%)	
PIR			0.36
Low	8356 (35.71%)	298 (34.89%)	
Medium	6288 (26.87%)	251 (29.39%)	
High	6594 (28.18%)	235 (27.52%)	
Unknown	2163 (9.24%)	70 (8.20%)	
Smoking status			< 0.01
Never	12,067 (51.57%)	370 (43.33%)	
Former	5385 (23.01%)	325 (38.06%)	
Now	4559 (19.48%)	157 (18.38%)	
Unknown	1390 (5.94%)	2 (0.23%)	
Drinking status			< 0.01
Never	2930 (12.52%)	139 (16.28%)	
Former	3375 (14.42%)	281 (32.90%)	
Mild	6762 (28.90%)	258 (30.21%)	
Moderate	3027 (12.94%)	66 (7.73%)	
Heavy	4150 (17.73%)	75 (8.78%)	
Unknown	3157 (13.49%)	35 (4.10%)	
Hypertension			0.34
No	12,626 (53.95%)	452 (52.93%)	
Yes	5347 (22.85%)	186 (21.78%)	
Unknown	5428 (23.20%)	216 (25.29%)	
DM			< 0.01
No	19,678 (84.09%)	717 (83.96%)	
Yes	2289 (9.78%)	65 (7.61%)	
Unknown	1434 (6.13%)	72 (8.43%)	

Table 1 (continued)

Characteristic	Alive	Death	P value
Hyperlipidemia			< 0.01
No	6825 (29.17%)	128 (14.99%)	
Yes	16,576 (70.83%)	726 (85.01%)	
CHD			< 0.01
No	20,829 (89.01%)	706 (82.67%)	
Yes	773 (3.30%)	135 (15.81%)	
Unknown	1799 (7.69%)	13 (1.52%)	
Stroke			< 0.01
No	20,976 (89.64%)	747 (87.47%)	
Yes	693 (2.96%)	103 (12.06%)	
Unknown	1732 (7.40%)	4 (0.47%)	

WC waist circumference, BMI body mass index, TyG triglyceride glucose index, TyG-WC TyG-waist circumference index, TyG-WHtR TyG-waist-to-height ratio index, TyG-WWI triglyceride glucose weight adjusted waist index, PIR poverty income ratio, DM diabetes mellitus, CHD coronary heart disease

considering the significant impact of age on mortality, models integrating TyG-related parameters with age were constructed and analyzed accordingly. To account for the potential influence of time on predictive accuracy, time-dependent ROC analyses were additionally performed, with ROC curves constructed at distinct time points (2, 5, 15, and 20 years). The incremental AUC and continuous NRI of the TyG-WWI relative to traditional TyG-related indices in predicting cardiovascular mortality were calculated [53]. Finally, considering the significant interaction with age, subgroup ROC analyses were further conducted within age-stratified populations.

To further quantify the absolute clinical significance of the associations between TyG-related parameters and cardiovascular death, we calculated the absolute rate differences in cardiovascular mortality between groups stratified by TyG indices. Specifically, thresholds for each TyG index (TyG, TyG-WC, TyG-WHtR, TyG-WWI) were determined via traditional ROC analysis to optimize the discriminative ability for cardiovascular death. On the basis of these thresholds, participants were categorized into low and high groups for each index. The absolute rate difference was computed as the difference in the incidence rate of cardiovascular death between the high and low groups, with 95% CIs estimated via the normal approximation method.

Given that adjusted covariates might influence the association between the TyG-WWI and cardiovascular mortality, prespecified subgroup analyses were conducted on the variables included in Model II to further evaluate potential effect modifications, aiming to clarify the possible interaction in the relationship between the TyG-WWI and cardiovascular mortality. The subgroup variables analyzed included age, sex, educational attainment, marital status, PIR, drinking status, smoking status, CHD, DM, hyperlipidemia, and hypertension history. Considering that conducting analyses across multiple subgroups may increase the risk of type I errors, multiple

Bonferroni corrections were applied with reference to previous guidelines for reporting subgroup analyses [54], resulting in a corrected significance threshold of $\alpha = 0.0042$.

Given the potential influence of evolving cardiovascular risk factors and treatment strategies over the extended study period (1999–2018), several supplementary analyses were conducted. First, stratified analyses by time period (1999–2008 and 2009–2018) were performed to examine the stability of the associations between the TyG-WWI and cardiovascular mortality across these intervals. Additionally, considering that lipid-lowering medication use might attenuate the relationship between triglycerides (a component of TyG) and cardiovascular risk, we analyzed temporal trends in lipid-lowering drug utilization rates and cardiovascular death incidence across the two time periods (1999–2008 vs. 2009–2018). Lipid-lowering drug use was also included as a covariate in the aforementioned time-stratified analyses. Furthermore, a landmark analysis was conducted to evaluate the stability of the prognostic value of the TyG-WWI over time, with the landmark time point set at 10 years. Specifically, participants were categorized into high- and low-TyG-WWI groups according to the optimal cutoff value derived from conventional ROC curve analysis. Survival differences between the two groups were then compared at time points both before and after the 10-year landmark throughout the follow-up period.

To further validate the robustness of the study findings, we conducted several supplementary sensitivity analyses in addition to multiple imputation for missing data. First, to address potential reverse causality (i.e., unrecognized subclinical diseases at baseline might concurrently influence TyG-WWI levels and short-term mortality risk), we excluded participants who died within 2 years of follow-up initiation and replicated the Cox regression analyses reported in Table 2 using this restricted dataset to examine result stability. Second, given that redefining missing

Table 2 Cox regression analyses for the associations between TyG, TyG-WC, TyG-WHtR, and TyG-WWI and cardiovascular mortality in different models

Exposure	No of deaths/No	HR (95%CI)		
		Crude model	Model I	Model II
TyG (per SD)	854/24255	1.49 (1.38,1.60)	1.24 (1.12,1.37)	1.24 (1.12, 1.38)
TyG-WC (per SD)	854/24255	1.63 (1.52,1.74)	1.39 (1.27,1.53)	1.39 (1.27, 1.53)
TyG-WHtR (per SD)	854/24255	1.73 (1.62,1.84)	1.43 (1.31,1.57)	1.43 (1.30, 1.58)
TyG-WWI (per SD)	854/24255	2.09 (1.94,2.26)	1.44 (1.31,1.58)	1.45 (1.31, 1.60)
TyG (quartiles)				
Q1	91/6063	1.0(Ref)	1.0(Ref)	1.0(Ref)
Q2	187/6063	1.72 (1.25,2.38)	0.89 (0.65,1.21)	0.93 (0.68, 1.26)
Q3	241/6061	2.38 (1.76,3.21)	0.96 (0.73,1.28)	0.97 (0.71, 1.33)
Q4	335/6068	3.54 (2.62,4.78)	1.30 (0.96,1.75)	1.33 (0.96, 1.83)
p-trend		< 0.01	< 0.01	0.01
TyG-WC (quartiles)				
Q1	85/6064	1.0(Ref)	1.0(Ref)	1.0(Ref)
Q2	196/6063	2.43 (1.77,3.33)	1.17 (0.85,1.61)	1.16 (0.84, 1.62)
Q3	240/6064	2.99 (2.25,3.97)	1.07 (0.81,1.43)	1.07 (0.78, 1.45)
Q4	333/6064	5.12 (3.84,6.81)	1.92 (1.45,2.55)	1.90 (1.40, 2.58)
p-trend		< 0.01	< 0.01	< 0.01
TyG-WHtR (quartiles)				
Q1	77/6064	1.0(Ref)	1.0(Ref)	1.0(Ref)
Q2	203/6063	2.88 (2.12,3.91)	1.19 (0.89,1.60)	1.21 (0.90, 1.63)
Q3	247/6064	3.46 (2.61,4.59)	1.14 (0.87,1.50)	1.17 (0.89, 1.53)
Q4	327/6064	6.32 (4.66,8.57)	1.98 (1.48,2.67)	2.01 (1.50, 2.69)
p-trend		< 0.01	< 0.01	< 0.01
TyG-WWI (quartiles)				
Q1	43/6064	1.0(Ref)	1.0(Ref)	1.0(Ref)
Q2	148/6063	3.86 (2.49,5.99)	1.65 (1.07, 2.56)	1.73 (1.10, 2.72)
Q3	259/6064	7.30 (4.79,11.13)	1.84 (1.21, 2.80)	1.96 (1.25, 3.09)
Q4	404/6064	14.69 (9.70,22.24)	2.78 (1.83, 4.23)	2.95 (1.85, 4.70)
p-trend		< 0.01	< 0.01	< 0.01

The crude model was adjusted for none

Model I adjusts for age and sex

Model II was adjusted for age, sex, ethnicity, education, marital status, PIR, drinking status, smoking status, CHD, DM, hyperlipidemia and hypertension history

HR hazard ratio, SD standard deviation. Other abbreviations are listed in Table 1

categorical variables alone may not fully mitigate the impact of data completeness on outcomes, we repeated the primary analysis using the original complete dataset after excluding all participants with missing values for categorical variables. Third, considering the potential influence of baseline cardiovascular disease on mortality outcomes (given that the risk pathway for death in this population may differ from that in individuals without underlying diseases), we excluded participants with baseline cardiovascular disease and reperformed Cox regression analysis to verify the independence of the observed associations. Finally, to account for the impact of potential unmeasured confounding factors, we calculated the E value via the E value package in R software [55], which quantifies the potential influence of unmeasured confounders on the association between the TyG-WWI and cardiovascular mortality risk.

Results

Baseline characteristics of the study participants

Table 1 presents the baseline characteristics of participants stratified by cardiovascular death status. A total of 23,401 participants remained alive, while 854 died from cardiovascular causes. Significant differences were observed in multiple key variables between the two groups ($P < 0.01$). The deceased participants were older (67.62 ± 0.69 vs. 45.61 ± 0.22 years) and had higher BMIs (29.49 ± 0.29 vs. 28.55 ± 0.08 kg/m²) and waist circumferences (104.22 ± 0.68 vs. 97.84 ± 0.20 cm). All TyG-related indices were significantly elevated in the deceased group, with the TyG-WWI showing a notable difference (102.92 ± 0.46 vs. 93.75 ± 0.16 , $P < 0.01$). With respect to demographic factors, the deceased group had a greater proportion of males (59.60% vs. 48.29%) and non-Hispanic Whites (60.77% vs. 42.86%). Lifestyle factors also differed significantly: deceased participants had higher

rates of former smoking (38.06% vs. 23.01%) and former alcohol consumption (32.90% vs. 14.42%). Comorbidities such as hyperlipidemia, coronary heart disease (CHD), and stroke were more prevalent in the deceased group (all $P < 0.01$), whereas no significant differences were found in education, marital status, PIR, or hypertension ($P > 0.05$).

Additional analyses provided further insights into the data characteristics. Multiple normality tests (Supplementary Table 6) collectively indicated that TyG-related parameters exhibited a skewed distribution, which was further verified by Q-Q plots (Supplementary Fig. 2) and histograms (Supplementary Fig. 3). Moreover, the baseline characteristics of the included versus excluded participants (Supplementary Table 7) revealed that most variables had standardized differences $< 10\%$ (considered nonsignificant), with only drinking status (28.5%) and hyperlipidemia (16.7%) showing notable differences. Other variables, such as age (3.8%), height (6.1%), BMI (2.8%), TyG (9.9%), ethnicity (4.4%), education (4.3%), marital status (6.9%), PIR (7.5%), smoking status (2.9%), hypertension (4.5%), DM (7.6%), CHD (2.0%), and stroke (3.9%), had standardized differences within the nonsignificant range, suggesting good comparability between the two groups.

Association analysis between TyG, TyG-WC, TyG-WHtR, and TyG-WWI and cardiovascular mortality

Cox regression analyses (Table 2) demonstrated that after adjustment for potential confounders in Model II, each standard deviation increase in the TyG-WWI was associated with a 45% higher risk of cardiovascular mortality (HR = 1.45, 95% CI 1.31–1.60), which was stronger than the corresponding associations for TyG (HR = 1.24, 95% CI 1.12–1.38), TyG-WC (HR = 1.39, 95% CI 1.27–1.53), and TyG-WHtR (HR = 1.43, 95% CI 1.30–1.58). When stratified by quartiles, the TyG-WWI exhibited a clear dose-response relationship (p -trend < 0.01) in Model II, with the highest quartile showing a 2.95-fold increased risk compared with the lowest quartile (95% CI 1.85–4.70). Although other TyG-related indices also presented significant p -trend values (< 0.01), their effect sizes across quartiles suggested potential nonlinear associations (e.g., nonmonotonic patterns in intermediate quartiles). Competing risk analyses (Supplementary Table 8), which accounted for deaths from other causes as competing events, revealed that while effect sizes were slightly reduced compared with the Cox regression results, the associations remained significant: each standard deviation increase in the TyG-WWI was associated with a 39% elevated risk of cardiovascular mortality (HR = 1.39, 95% CI 1.28–1.51), and the highest quartile of the TyG-WWI remained significantly associated with a 2.31-fold higher risk (95% CI 1.64–3.24) relative to the lowest quartile

(p -trend < 0.01). Similar patterns were observed for other TyG-related indices in competing risk analyses, with TyG-WHtR (HR = 1.34 per SD, 95% CI 1.24–1.45) and TyG-WC (HR = 1.31 per SD, 95% CI 1.21–1.42) showing weaker but still significant associations and TyG exhibiting the weakest association (HR = 1.22 per SD, 95% CI 1.13–1.32). Stratification by ROC-derived thresholds (Supplementary Table 9) further revealed that individuals in the high TyG-WWI group had a 69% greater risk of cardiovascular death (HR = 1.69, 95% CI 1.41–2.04) than did those in the low TyG-WWI group, with an absolute rate difference of 0.44% (95% CI 0.15–0.73), which was greater than the absolute risk differences observed for other TyG-related indices. These findings collectively indicate that the TyG-WWI has a robust and consistent association with cardiovascular mortality, with stronger risk stratification ability than other TyG-based indices.

Temporal stability of associations between TyG-related indices and cardiovascular mortality across time periods

Stratified analyses by time period (1999–2008 and 2009–2018) revealed that the associations between TyG-related indices (including the TyG-WWI) and cardiovascular mortality remained relatively consistent across the two intervals (Supplementary Table 13). Notably, although the magnitude of associations for all TyG-related indices with cardiovascular death was slightly stronger in 2009–2018 than in 1999–2008, no statistically significant interactions were observed (all P -interaction > 0.05), indicating minimal temporal variation in these associations. The landmark analysis further corroborated this stability, as survival probability differences between the low- and high-TyG-WWI groups persisted significantly ($p < 0.001$) across both the pre- and post-120-month intervals (Supplementary Fig. 4). Concurrently, temporal trends in therapeutic interventions and outcomes revealed that the prevalence of lipid-lowering drug use increased from 12.98 to 20.18%, whereas the incidence of cardiovascular death decreased from 5.52 to 1.61% over the same periods (Supplementary Table 14). Importantly, the inclusion of lipid-lowering drug use as a covariate in the models did not substantially alter the effect sizes of TyG-related indices, with the interaction results remaining consistent with those of the primary analyses.

Dose-response relationships between TyG, TyG-WC, TyG-WHtR, and TyG-WWI and cardiovascular mortality

Restricted cubic spline (RCS) plots (Fig. 2) and two-piecewise linear regression models were used to explore potential nonlinear associations between TyG-related indices and cardiovascular mortality, with all indices showing significant overall associations with cardiovascular death (P -overall < 0.001). Specifically, RCS analyses revealed distinct association patterns: TyG exhibited

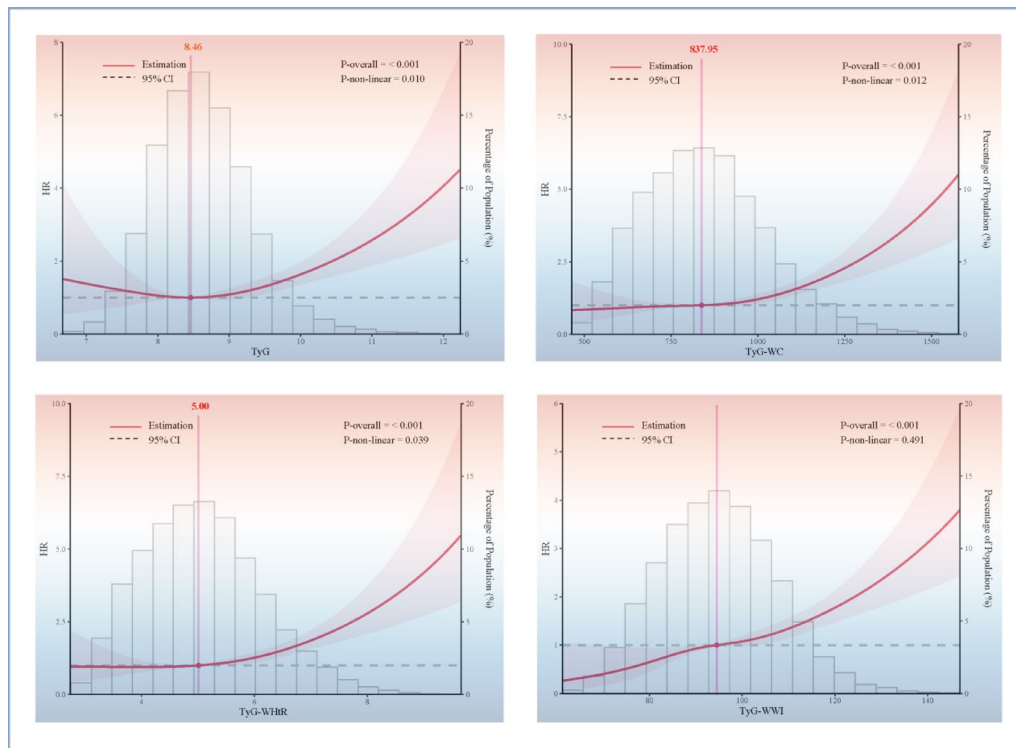


Fig. 2 Dose–response relationships between the TyG, TyG-WC, TyG-WHtR, and TyG-WWI and cardiovascular mortality. All abbreviations are listed in Table 1. Restricted cubic splines were adjusted for age, sex, ethnicity, education, marital status, PIR, drinking status, smoking status, CHD, DM, hyperlipidemia and hypertension history

a U-shaped nonlinear association ($P\text{-nonlinear} < 0.05$), whereas TyG-WC and TyG-WHtR showed L-shaped nonlinear associations (both $P\text{-nonlinear} < 0.05$). In contrast, the TyG-WWI demonstrated a linear positive association with cardiovascular mortality ($P\text{-non-linear} = 0.491$). Two-piecewise linear regression further quantified these associations by identifying inflection points (Supplementary Tables 10–12). For TyG, the inflection point was 8.46: below this threshold, TyG was not significantly associated with cardiovascular death ($HR = 0.81$, 95% CI 0.50–1.32, $P = 0.39$), whereas above 8.46, each unit increase in TyG was associated with a 32% higher risk ($HR = 1.32$, 95% CI 1.18–1.49, $P < 0.01$). For TyG-WC, the inflection point was 837.95: no significant association was observed below this threshold ($HR = 1.10$, 95% CI 0.69–1.73, $P = 0.70$), but above 837.95, each unit increase was linked to a 61% elevated risk ($HR = 1.61$, 95% CI 1.42–1.83, $P < 0.01$). Similarly, TyG-WHtR had an inflection point of 5.00: below 5.00, there was no significant association ($HR = 1.35$, 95% CI 0.82–2.23, $P = 0.24$), whereas above 5.00, each unit increase was associated with a 63% higher risk ($HR = 1.63$, 95% CI 1.41–1.89, $P < 0.01$). These findings were further validated by log-likelihood ratio tests, with all P values < 0.05 (0.040 for TyG, 0.024 for TyG-WC, and 0.027 for TyG-WHtR),

confirming the suitability of the piecewise models for capturing nonlinear associations.

Comparison of the predictive value of the TyG-WWI and TyG-related parameters for cardiovascular mortality

We evaluated the predictive performance of TyG-related indices for cardiovascular mortality via both conventional and time-dependent receiver operating characteristic (ROC) analyses. Conventional ROC analysis (Fig. 3 and Table 3) revealed that the TyG-WWI had the highest AUC (0.694, 95% CI 0.678–0.710), which was significantly greater than that of TyG (0.627, 95% CI 0.609–0.645), TyG-WC (0.630, 95% CI 0.612–0.647), and TyG-WHtR (0.635, 95% CI 0.618–0.652), as indicated by Delong's test with Bonferroni correction (all $P < 0.01$). The optimal threshold for the TyG-WWI was 96.5235, with a specificity of 0.5729 and sensitivity of 0.7201. Notably, while the TyG-WWI had a slightly greater AUC than the WWI did in the overall population (0.694 vs. 0.693), the difference was not statistically significant ($P = 0.87$; Supplementary Table 15). Time-dependent ROC analysis (Fig. 4) further confirmed the superiority of the TyG-WWI, with longitudinal AUCs ranging from 0.706 to 0.751 across different follow-up time points, which were consistently greater than those of TyG (AUC range: 0.595–0.661), TyG-WC

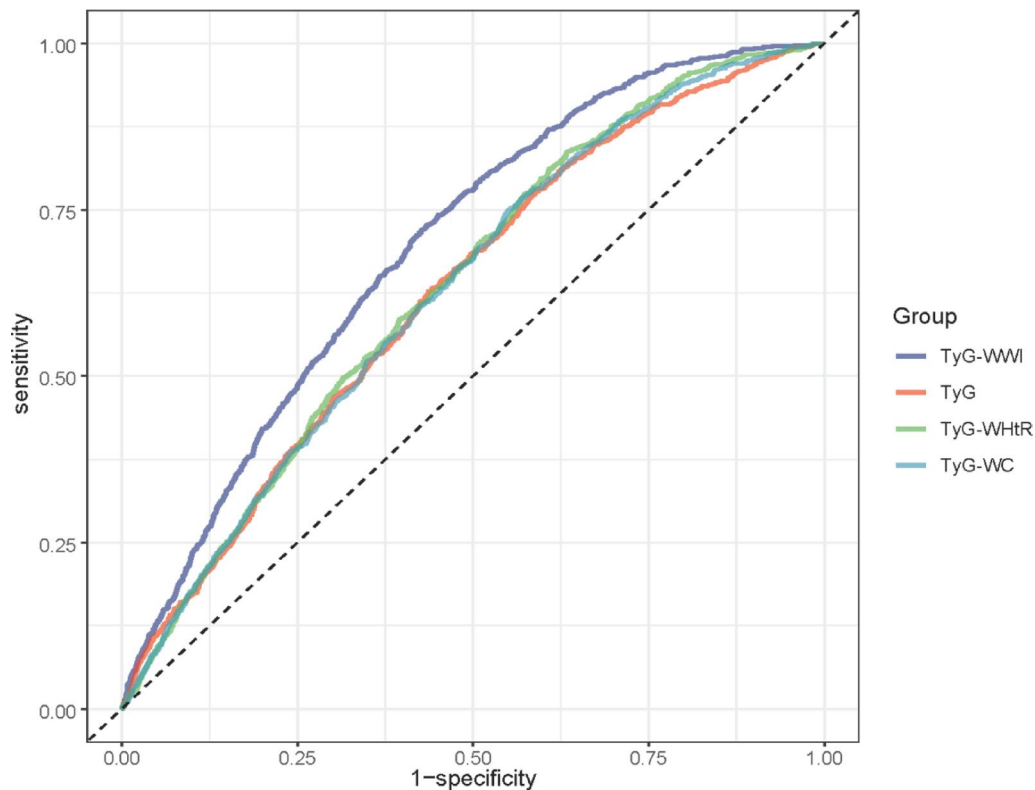


Fig. 3 ROC curves of the TyG, TyG-WC, TyG-WHtR, and TyG-WWI in predicting cardiovascular mortality. The optimal threshold for the ROC curve is determined by the Youden index. ROC receiver operating characteristic. Other abbreviations are listed in Table 1

Table 3 ROC analysis comparing the ability of the TyG-WWI and TyG-related parameters to predict cardiovascular mortality

Test	AUC (95%CI)	Best threshold	Specificity	Sensitivity	P†
TyG	0.627 (0.609–0.645)	8.6752	0.5629	0.6276	< 0.01
TyG-WC	0.630 (0.612–0.647)	811.1431	0.4496	0.7494	< 0.01
TyG-WHtR	0.635 (0.618–0.652)	4.6952	0.3919	0.6952	< 0.01
TyG-WWI	0.694 (0.678–0.710)	96.5235	0.5729	0.7201	–

ROC receiver operating characteristic, AUC area under the curve. Other abbreviations are listed in Table 1

†Delong’s test *p* value comparing each index with TyG-WWI, with Bonferroni correction applied for multiple comparisons (adjusted significance threshold $\alpha=0.017$)

(AUC range: 0.613–0.711), and TyG-WHtR (AUC range: 0.622–0.701) throughout the follow-up period.

In the age-stratified analyses, all indices presented higher AUC values in the ≤ 60 years subgroup than in the > 60 years subgroup. In the age ≤ 60 subgroup, the TyG-WWI had the highest AUC (0.703, 95% CI 0.664–0.742), which was significantly greater than that of the

WWI ($P < 0.001$) after Bonferroni correction, although no significant differences were observed compared with the other TyG-related indices (all $P > 0.012$). In the age > 60 subgroup, the TyG-WWI still had the highest AUC (0.553, 95% CI 0.530–0.575), with a significant difference only compared with the TyG-WHtR ($P < 0.01$), while no significant differences were found in the other indices (all $P > 0.012$). Thus, although the TyG-WWI had the numerically highest AUC in both age subgroups, statistical significance was limited to specific comparisons, reinforcing its superior predictive performance, particularly in younger populations.

Incremental predictive value of the TyG-WWI compared with traditional TyG parameters

The TyG-WWI demonstrated significant incremental predictive value over traditional TyG-related parameters, as supported by multiple lines of evidence. Time-dependent analyses (Supplementary Table 16) revealed that across key follow-up time points (24, 60, 120, 180, and 240 months), the TyG-WWI consistently achieved higher AUCs (ranging from 0.714 to 0.753) than did TyG (0.612–0.659), TyG-WC (0.622–0.703), and TyG-WHtR (0.634–0.701), with substantial delta values. The AUC increased (0.050–0.111) at each time point. Concurrently, continuous net reclassification improvement (NRI)

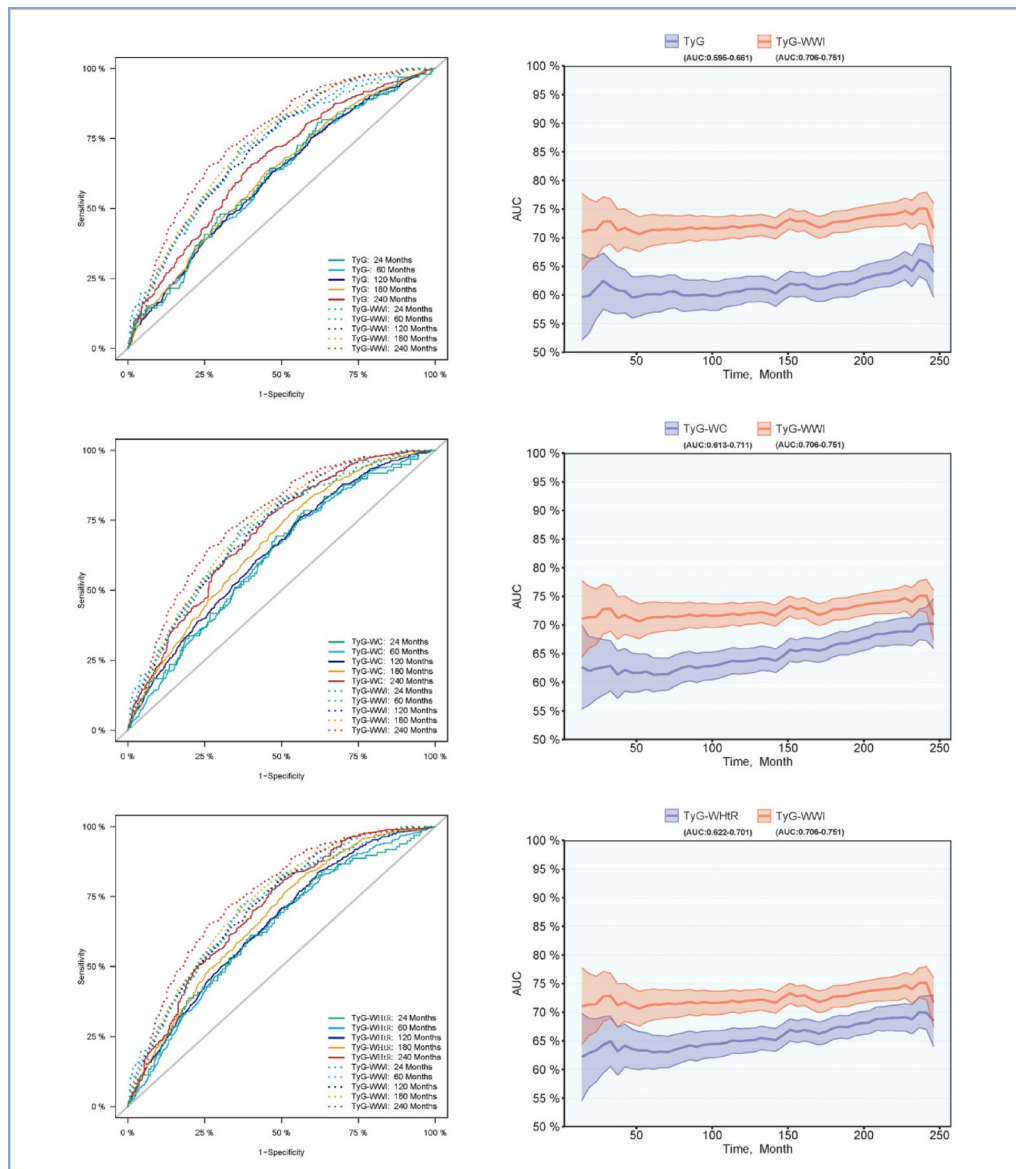


Fig. 4 Time-dependent predictive performance comparison: ROC curves and longitudinal AUC trends of TyG, TyG-WC, TyG-WHtR versus TyG-WWI. ROC receiver operating characteristic; AUC area under the curve. Other abbreviations are listed in Table 1

analyses confirmed this superiority: TyG-WWI showed significant NRI increases versus TyG (35.5–41.4%), TyG-WC (26.5–34.6%), and TyG-WHtR (32.1–35.3%) across all time points (all $P < 0.001$ after Bonferroni correction). Furthermore, calibration assessments (Supplementary Fig. 5) indicated that all models integrating TyG-related parameters with age exhibited good calibration, as evidenced by Hosmer–Lemeshow test P values > 0.05 ; notably, the TyG-WWI + Age model showed the closest alignment with the ideal diagonal ($P = 0.623$), outperforming TyG + Age ($P = 0.581$), TyG-WHtR + Age ($P = 0.305$), and TyG-WC + Age ($P = 0.082$). Complementing these findings, overall NRI analyses (Supplementary Table 17) quantitatively confirmed the incremental value

of TyG-WWI, with NRI values of 10.4% (7.4–13.3%) versus TyG, 9.4% (6.5–12.3%) versus TyG-WC, and 9.1% (6.5–12.7%) versus TyG-WHtR.

Subgroup analysis

Stratified analyses were performed to explore the consistency of associations between the TyG-WWI and cardiovascular mortality across different subgroups (Table 4), with Bonferroni correction applied to the significance threshold for interaction tests (adjusted $\alpha = 0.0042$). Overall, the TyG-WWI was significantly associated with an elevated risk of cardiovascular mortality in most subgroups, with adjusted hazard ratios (HRs) ranging from 1.21 to 1.65. Notably, a significant

Table 4 Stratified analyses of the associations between the TyG-WVI and cardiovascular mortality

Subgroup	No. of participants	Adjusted HR (95%CI)	P-interaction
Age, years			< 0.001
≤ 60	12,445	2.03 (1.78, 2.32)	
> 60	11,810	1.35 (1.24, 1.48)	
Sex			0.38
Female	12,445	1.43 (1.28, 1.60)	
Male	11,810	1.34 (1.20, 1.49)	
Ethnicity			0.31
Non-Hispanic White	10,549	1.33 (1.20, 1.48)	
Other races	13,706	1.23 (1.09, 1.39)	
Education			0.19
Below high school	5814	1.25 (1.07, 1.46)	
High school or equivalent	7042	1.33 (1.15, 1.53)	
Above high school	7311	1.44 (1.25, 1.66)	
Unknown	4088	1.59 (1.33, 1.89)	
Marital status			0.59
Married/cohabiting	7810	1.35 (1.17, 1.55)	
Widowed/divorced/separated	2985	1.50 (1.21, 1.86)	
Never married	3816	1.28 (1.07, 1.52)	
Unknown	9644	1.43 (1.27, 1.62)	
PIR			0.96
Low	8654	1.38 (1.22, 1.57)	
Medium	6539	1.36 (1.18, 1.58)	
High	6829	1.42 (1.23, 1.63)	
Unknown	2233	1.31 (0.98, 1.75)	
Drinking status			0.87
Never/past	6725	1.37 (1.22, 1.52)	
Current	14,338	1.40 (1.24, 1.57)	
Unknown	3192	1.50 (1.05, 2.14)	
Smoking status			0.03
Never/past	18,147	1.46 (1.33, 1.59)	
Current	4716	1.14 (0.96, 1.35)	
Unknown	1392	0.95 (0.25, 3.61)	
CHD			0.65
No	21,535	1.40 (1.29, 1.53)	
Yes	908	1.27 (1.03, 1.56)	
Unknown	1812	1.29 (0.61, 2.70)	
DM			0.33
No	20,395	1.36 (1.24, 1.48)	
Yes	2354	1.65 (1.29, 2.12)	
Unknown	1506	1.39 (1.09, 1.77)	
Hyperlipidemia			0.69
No	6953	1.32 (1.05, 1.67)	
Yes	17,302	1.39 (1.28, 1.51)	
Hypertension			0.08
No	13,078	1.28 (1.15, 1.43)	
Yes	5533	1.46 (1.25, 1.71)	
Unknown	5644	1.56 (1.33, 1.82)	

All abbreviations are listed in Table 1. The P-interaction was compared with the α value after Bonferroni correction to reduce the risk of Type I error, and the corrected $\alpha = 0.0042$

interaction was observed by age (P-interaction < 0.001), where the association was stronger in participants aged ≤ 60 years (HR = 2.03, 95% CI 1.78–2.32) than in those aged > 60 years (HR = 1.35, 95% CI 1.24–1.48). In

contrast, no significant interactions were detected for the other subgroups, as the interaction P values all exceeded the adjusted significance threshold. Moreover, the associations remained directionally consistent within these

Table 5 Sensitivity analysis of the associations between TyG, TyG-WC, TyG-WHtR, and TyG-WWI and cardiovascular mortality

	HR (95% CI)			
	Sensitivity-1	Sensitivity-2	Sensitivity-3	Sensitivity-4
TyG (per SD)	1.24 (1.11, 1.37)	1.24 (1.12, 1.37)	1.24 (1.11, 1.37)	1.24 (1.12, 1.38)
TyG-WC (per SD)	1.39 (1.26, 1.53)	1.39 (1.26, 1.53)	1.38 (1.25, 1.53)	1.39 (1.26, 1.53)
TyG-WHtR (per SD)	1.42 (1.29, 1.57)	1.42 (1.29, 1.57)	1.42 (1.29, 1.57)	1.42 (1.29, 1.57)
TyG-WWI (per SD)	1.44 (1.31, 1.59)	1.44 (1.31, 1.59)	1.44 (1.30, 1.59)	1.44 (1.31, 1.59)
TyG (quartiles)				
Q1	1.0(Ref)	1.0(Ref)	1.0(Ref)	1.0(Ref)
Q2	0.94 (0.69, 1.27)	0.94 (0.69, 1.28)	0.94 (0.69, 1.28)	0.94 (0.69, 1.28)
Q3	0.98 (0.72, 1.34)	0.98 (0.72, 1.34)	0.98 (0.72, 1.34)	0.98 (0.72, 1.34)
Q4	1.32 (0.96, 1.82)	1.33 (0.97, 1.84)	1.32 (0.96, 1.83)	1.33 (0.96, 1.84)
p-trend	0.0127	0.0119	0.0141	0.0130
TyG-WC (quartiles)				
Q1	1.0(Ref)	1.0(Ref)	1.0(Ref)	1.0(Ref)
Q2	1.16 (0.83, 1.62)	1.17 (0.84, 1.63)	1.17 (0.85, 1.63)	1.17 (0.84, 1.63)
Q3	1.09 (0.80, 1.47)	1.09 (0.80, 1.47)	1.09 (0.81, 1.48)	1.09 (0.80, 1.47)
Q4	1.88 (1.38, 2.55)	1.89 (1.40, 2.56)	1.88 (1.39, 2.55)	1.89 (1.39, 2.57)
p-trend	< 0.01	< 0.01	< 0.01	< 0.01
TyG-WHtR (quartiles)				
Q1	1.0(Ref)	1.0(Ref)	1.0(Ref)	1.0(Ref)
Q2	1.22 (0.92, 1.64)	1.22 (0.91, 1.64)	1.23 (0.92, 1.64)	1.23 (0.92, 1.64)
Q3	1.18 (0.91, 1.55)	1.18 (0.90, 1.55)	1.18 (0.90, 1.54)	1.18 (0.90, 1.54)
Q4	2.00 (1.49, 2.68)	2.00 (1.49, 2.69)	2.00 (1.49, 2.68)	2.00 (1.49, 2.69)
p-trend	< 0.01	< 0.01	< 0.01	< 0.01
TyG-WWI (quartiles)				
Q1	1.0(Ref)	1.0(Ref)	1.0(Ref)	1.0(Ref)
Q2	1.75 (1.11, 2.76)	1.75 (1.11, 2.76)	1.75 (1.11, 2.75)	1.76 (1.12, 2.77)
Q3	2.00 (1.27, 3.13)	2.00 (1.28, 3.13)	2.00 (1.27, 3.12)	2.01 (1.28, 3.15)
Q4	2.98 (1.87, 4.73)	2.99 (1.88, 4.76)	2.97 (1.87, 4.71)	2.99 (1.88, 4.76)
p-trend	< 0.01	< 0.01	< 0.01	< 0.01

HR hazard ratio, SD standard deviation. Other abbreviations are listed in Table 1

Sensitivity-1 to sensitivity-4 were repeated analyses using data after multiple imputations of missing covariates

subgroups, with the TyG-WWI consistently linked to a higher risk of cardiovascular mortality across various demographic, socioeconomic, and clinical strata. These findings indicate that the association between the TyG-WWI and cardiovascular mortality is generally consistent across most subgroups, with a notably stronger effect observed in younger individuals.

Sensitivity analysis

Sensitivity analyses were conducted to validate the robustness of the associations between TyG-related indices and cardiovascular mortality. In multiple imputation-based sensitivity analyses (Table 5), the adjusted hazard ratios (HRs) for the TyG-WWI per standard deviation increment remained consistent across the four imputed datasets (1.44, 95% CI 1.30–1.59), with similar stabilities observed for TyG, TyG-WC, and TyG-WHtR. Quartile-based analyses further confirmed a persistent dose-response relationship for the TyG-WWI (all p values < 0.01), mirroring the primary findings.

Additional sensitivity analyses (Supplementary Table 18) yielded consistent results: excluding participants who died within 2 years of follow-up (sensitivity-1), removing all missing data (sensitivity-2), and specifically, sensitivity-3, which excluded participants with baseline cardiovascular disease, revealed that the TyG-WWI retained significant associations with cardiovascular mortality (HR per standard deviation (SD): 1.43–1.46). Notably, in the subgroup without baseline cardiovascular disease, ROC analysis (Supplementary Table 19) demonstrated that the TyG-WWI still had the highest AUC (0.661, 95% CI 0.632–0.691), which was significantly greater than that of TyG (0.610, 95% CI 0.579–0.641), TyG-WC (0.607, 95% CI 0.576–0.637), and TyG-WHtR (0.615, 95% CI 0.584–0.645) according to Delong's test (all p < 0.01).

Furthermore, the E value quantifying the impact of potential confounding factors was 2.26; this indicates that an unmeasured confounder would need to be associated with both the TyG-WWI and cardiovascular death by approximately 2.3-fold to render the lower bound of the

HR confidence interval for the association between the TyG-WWI and cardiovascular death risk < 1 . Given that many known confounders have already been adjusted for, it is unlikely that a single unmeasured residual confounder could achieve such a magnitude of association.

Discussion

In this representative cohort study based on a general population sample, we conducted the first assessment of the association between the TyG-WWI and cardiovascular mortality in the general population, with a specific focus on its incremental predictive value compared with traditional TyG-related parameters. The results showed that after adjusting for multiple confounding factors, each one-standard deviation increase in the TyG-WWI was associated with a 45% higher risk of cardiovascular mortality (HR = 1.45 (1.31–1.60)), and this risk was greater than that observed for TyG, TyG-WC, and TyG-WHtR. Additionally, both traditional and time-dependent ROC curves confirmed that the TyG-WWI had greater accuracy in predicting cardiovascular mortality. Importantly, NRI analyses further confirmed its incremental predictive value: compared with TyG, TyG-WC, and TyG-WHtR, the NRI values for the TyG-WWI were 10.4%, 9.4%, and 9.1%, respectively. These findings collectively demonstrate that the TyG-WWI is a reliable predictor of cardiovascular mortality and has significant incremental predictive value over traditional TyG-related parameters, supporting its potential as a more promising risk assessment tool.

Comparison with prior studies

Previous epidemiological studies have explored the associations between various TyG-related parameters and cardiovascular mortality. While most investigations support a link between TyG-related indices and cardiovascular mortality, evidence remains inconsistent across diverse populations. For example, in two studies utilizing data from the UK Biobank [29] and the US NHANES [56], TyG showed no significant association with cardiovascular mortality in individuals with metabolic dysfunction-associated steatotic liver disease (MASLD). Consistently, a cohort study involving 10,734 participants with metabolic syndrome yielded similar findings [57]. In contrast, TyG-WC and TyG-WHtR have demonstrated significant correlations with cardiovascular outcomes across multiple populations [29–31, 56–60]. These discrepant results may be attributed to variations in study design and population characteristics. In another general population study, TyG, TyG-WC, and TyG-WHtR all exhibited significant positive correlations with cardiovascular mortality [60]. Our findings align with these previous observations: even after adjusting for demographic characteristics, lifestyle factors, and baseline

comorbidities, we consistently identified associations between TyG-related parameters and cardiovascular mortality. When accounting for the competing risk of noncardiovascular death, the strength of the association between each index and cardiovascular mortality was slightly attenuated, but the direction remained consistent with the Cox regression results, indicating that their predictive value for cardiovascular mortality remains robust after adjusting for competing risks. Notably, our study provides novel insight: the newly proposed TyG-WWI was significantly and linearly associated with cardiovascular mortality in the general population. Furthermore, after all the parameters were standardized, the TyG-WWI outperformed the TyG in assessing cardiovascular mortality risk and performed comparably to the TyG-WC and TyG-WHtR. This finding is consistent with prior evidence highlighting that TyG-WC and TyG-WHtR outperform TyG alone in evaluating cardiovascular outcomes [29–31, 56–60]. Notably, the subgroup analyses in the present study revealed a significant age-modifying effect, with the association between the TyG-WWI and cardiovascular mortality being particularly pronounced in younger populations (≤ 60 years). This observation aligns with two previous studies [59, 60], which reported stronger associations between TyG-related indices and cardiovascular mortality in individuals under 65 years of age. These age-related differences may reflect the cumulative impact of metabolic disorders over time, suggesting that early identification of high-risk individuals via the TyG-WWI and timely intervention could help mitigate long-term cardiovascular mortality risk.

Controversy persists regarding which TyG-related index optimally predicts cardiovascular outcomes [24, 28–31]. In a study of patients with cardiometabolic diseases (CMDs) [30], Zhang et al. reported that TyG had greater predictive accuracy for cardiovascular mortality than TyG-WC and TyG-WHtR did. This result was corroborated in another study involving individuals with MASLD combined with prediabetes or diabetes [28]. Interestingly, an opposite finding was reported in another investigation of the same MASLD population, where TyG-WC and TyG-WHtR outperformed TyG in predicting cardiovascular mortality [29]. Such discrepancies may partially stem from population differences: in individuals with preexisting metabolic diseases, the incremental value of combining TyG with obesity indices could be diminished, as metabolic disorders themselves exert a strong influence on cardiovascular risk, potentially overshadowing the added value of composite indices. Additionally, variations in study design and covariate adjustment may have contributed to this heterogeneity. In the present study, we compared the newly proposed TyG-WWI with traditional TyG-related parameters. The results from both conventional ROC analysis and

time-dependent ROC analysis consistently demonstrated that the TyG-WWI outperformed the TyG in predicting cardiovascular mortality, with a significantly larger AUC. When TyG was combined with different obesity indicators, TyG-WWI also showed superior performance compared with TyG-WC and TyG-WHtR, which is consistent with previous findings in a population with MASLD combined with prediabetes or diabetes [28]. Similarly, further NRI analysis revealed that the TyG-WWI achieved significant net reclassification gains compared with TyG, TyG-WC, and TyG-WHtR in both traditional and time-dependent ROC analyses, indicating its superior ability to correctly stratify individuals' cardiovascular death risk. These findings emphasize that the TyG-WWI offers significant incremental predictive value over traditional TyG-related parameters. Although both TyG-WC and TyG-WHtR incorporate obesity-related anthropometric measurements, the integration of central and general obesity in the TyG-WWI appears to yield a more optimized predictive model. A key reason for this superiority may lie in the limitations of single obesity parameters: WC and WHtR, which do not account for weight, fail to comprehensively assess fat distribution, leading to variable performance when combined with TyG. In contrast, the WWI integrates weight with waist circumference, providing a more holistic evaluation of adiposity. These results provide robust evidence supporting the application of the TyG-WWI in predicting cardiovascular mortality, indicating its ability to enhance risk prediction beyond traditional TyG-related parameters.

Several notable findings emerged from the current study. First, the discriminative performance of time-dependent ROC analysis across all time points ($AUC=0.706\text{--}0.751$) exceeded that of standard ROC analysis ($AUC=0.694$). This discrepancy may arise from differing handling of the temporal dimension: the standard ROC aggregates all cardiovascular mortality events over the entire follow-up period to reflect the overall performance of the TyG-WWI, whereas the time-dependent ROC focuses on specific time points, incorporating follow-up duration and censoring dynamics to more accurately capture its time-varying prognostic value. As follow-up has progressed, the predictive weight of metabolic and anthropometric information integrated by the TyG-WWI in assessing cumulative cardiovascular damage has increased, thereby enhancing its discriminative ability. Notably, our standard ROC analysis revealed relatively similar AUC values for TyG (0.627), TyG-WC (0.630), and TyG-WHtR (0.635), indicating comparable overall discriminative power in predicting cardiovascular mortality. This phenomenon may be attributed to the shared metabolic information they capture: TyG primarily reflects insulin resistance, whereas WC and WHtR focus on central obesity—a key correlate of metabolic

dysfunction. When combined with TyG, WC and WHtR may not add sufficient unique information to substantially improve discriminative power, as central obesity and insulin resistance often overlap in their associations with cardiovascular risk. Additionally, the limitations of single obesity parameters (WC and WHtR) contribute to this similarity: both indices assess central adiposity but neglect overall weight or body mass, restricting their ability to capture the full spectrum of adiposity-related cardiovascular risk and thus limiting their incremental value when combined with TyG. Furthermore, our study revealed that the AUC of the TyG-WWI for predicting cardiovascular mortality was slightly greater than that of the WWI alone, although without statistical significance. However, after age stratification, the AUCs of indices that combine the TyG index with obesity-related parameters (including the TyG-WWI) were significantly greater than those of the WWI. This finding indicates that the incremental predictive value of combining the TyG index with obesity indices is more pronounced in younger individuals. A plausible explanation is that cardiovascular risk in young people is more closely linked to interactions between early metabolic disorders (e.g., insulin resistance) and obesity—a synergistic effect not fully captured by a single obesity index (e.g., WWI). TyG sensitively reflects metabolic dysfunction, and its integration helps better identify potential risks arising from metabolism-obesity interactions. Given that young adults are a key target population for cardiovascular primary prevention, this age-specific incremental value positions the TyG-WWI as a more targeted tool for the early identification of high-risk young individuals.

The association between the TyG-WWI and cardiovascular mortality may be partially explained by the following mechanisms. First, insulin resistance—a core pathological feature of metabolic disorders and cardiovascular disease—is effectively captured by the TyG component of the TyG-WWI, which reflects impaired glucose and lipid metabolism. Mitochondrial dysfunction serves as a critical link between insulin resistance and cardiovascular disease progression [61–63]. In states of insulin resistance, mitochondrial oxidative phosphorylation in cardiac and vascular cells is impaired, leading to reduced ATP production and increased reactive oxygen species (ROS) generation. This not only exacerbates insulin resistance by disrupting intracellular signaling pathways but also induces endothelial dysfunction, vascular inflammation, and myocardial energy metabolism disorders [64–69], key processes driving the development of cardiovascular diseases such as atherosclerosis and heart failure, ultimately increasing the risk of cardiovascular mortality. Additionally, accumulating evidence highlights mitochondrial dysfunction as a significant contributor to cardiovascular disease progression [70], with improved

mitochondrial dysfunction and oxidative stress emerging as novel therapeutic targets for metabolic-related cardiovascular diseases [71–75]. Second, the central obesity reflected by the WWI component of the TyG-WWI amplifies mitochondrial dysfunction-induced cardiovascular damage through a synergistic effect. Ectopic fat deposition (e.g., visceral adiposity) associated with central obesity releases excessive free fatty acids and proinflammatory cytokines, which can directly inhibit mitochondrial respiratory chain complex activity, disrupt mitochondrial dynamics (e.g., imbalance between fission and fusion), and reduce the expression of key regulators of mitochondrial biogenesis (e.g., PGC-1 α) [76, 77]. This mitochondrial dysfunction, synergistically induced by central obesity and insulin resistance, may accelerate vascular wall lipid deposition, promote foam cell formation, and impair myocardial energy reserve capacity—ultimately increasing the risk of fatal cardiovascular events such as myocardial infarction and heart failure.

In our study, the TyG-WWI demonstrated greater predictive value for cardiovascular mortality than traditional TyG-related indices did. We hypothesize that this enhanced predictive ability stems from its comprehensive reflection of the synergistic interaction between insulin resistance and central obesity, as well as its ability to capture the dynamic fat distribution relative to total body weight. Specifically, elevated waist circumference in the context of excess weight may exacerbate insulin resistance by increasing adipose tissue-derived proinflammatory cytokines, impairing glucose metabolism, and promoting atherogenic dyslipidemia—core pathways linking TyG to cardiovascular risk. In contrast, traditional TyG-related indices (e.g., TyG, TyG-WC, TyG-WHtR) integrate insulin resistance and obesity-related information either singularly or partially, failing to fully capture such synergistic effects: TyG solely reflects insulin resistance; TyG-WC incorporates waist circumference but lacks weight-based adjustment for obesity severity, potentially overestimating or underestimating central obesity risk in individuals with varying body weights; TyG-WHtR normalizes waist circumference to height to account for body size differences but ignores the metabolic impact of overall fat burden reflected by weight, limiting its ability to precisely capture the synergy between central obesity and systemic metabolic status. For example, “metabolic mismatch” caused by excessive central fat in individuals with low total body weight may be underestimated by the WHtR, as its height standardization does not account for weight-dependent fat distribution patterns. In comparison, the WWI quantifies relative central obesity more accurately through the waist-to-weight ratio [36, 78]. Its weight adjustment accounts for body size differences, ensuring that identical waist circumferences carry distinct risk implications

for individuals with different weights—aligning with the physiological reality that cardiovascular risk arises from interactions among multiple anthropometric factors. This nuance enables WWI to capture obesity-related risk profiles that align with the metabolic phenotypic heterogeneity of the general population, which, when combined with TyG, more comprehensively captures the “insulin resistance-mitochondrial dysfunction-vascular damage” cascade. Mitochondrial dysfunction-induced abnormalities in apoptosis and autophagy [79, 80] may represent another critical link: impaired mitochondrial structure and function can activate apoptotic pathways, exacerbate oxidative stress and inflammation, and form a progressive pathological chain of “mitochondrial damage-tissue injury-cardiovascular events” [81, 82]. Elevated TyG-WWI values may indicate heightened activation of this pathological cascade, thereby more strongly predicting increased cardiovascular mortality risk. Collectively, these processes—from insulin resistance (captured by TyG) to mitochondrial dysfunction and subsequent progressive cardiovascular disease—partially explain the incremental predictive value of the TyG-WWI.

Clinical and public health implications

The findings of this study have meaningful implications for clinical practice. Currently, identifying individuals at high risk of cardiovascular mortality remains a critical challenge in clinical settings [83, 84], as traditional risk assessment tools often fail to capture the complex interplay between insulin resistance and obesity [85, 86]. As a more comprehensive predictive index, the TyG-WWI—a composite of the TyG index and WWI—exhibits incremental predictive value beyond existing TyG-based indices (TyG, TyG-WC, TyG-WHtR). Its advantages lie in two key aspects: first, it is derived from easily measurable parameters (triglycerides, glucose, weight, waist circumference) that require no specialized equipment, rendering it particularly feasible for resource-limited settings (e.g., primary care facilities with limited access to advanced diagnostic tools) where sophisticated cardiovascular risk assessment modalities are scarce. Second, its superior NRI and discriminative ability (AUC) support its potential role in refining risk stratification for adults at risk of cardiovascular death, especially in identifying individuals with moderately elevated risk, who might be overlooked by conventional assessments. This can assist clinicians in prioritizing lifestyle counseling or further diagnostic evaluations for such individuals, thereby facilitating more targeted risk management.

Over time, the widespread adoption of the TyG-WWI could contribute to more personalized preventive strategies, potentially reducing the burden of cardiovascular mortality through improved risk identification. As a valuable addition to existing risk assessment toolkits,

TyG-WWI can be readily implemented in various medical settings because of its reliance on routine clinical measurements (triglycerides, blood glucose, waist circumference, and body weight), imposing no additional diagnostic burden and thus exhibiting high applicability in routine clinical practice. By integrating the TyG-WWI, clinicians can more effectively identify high-risk patients and implement targeted preventive strategies and interventions, thereby enhancing the effectiveness of cardiovascular disease prevention.

Importantly, while the TyG-WWI shows promise in risk stratification and prognostic assessment for cardiovascular mortality—providing clinicians with valuable supplementary information—it should be regarded as a complementary tool rather than a standalone predictor. In fact, a single index with an AUC of ~ 0.694 cannot fully encapsulate the multifactorial complexity of cardiovascular mortality risk, which is consistent with our recognition that cardiovascular risk prediction models such as the Framingham risk score and ASCVD risk calculation [87–89], which have been validated across diverse populations, remain cornerstones of cardiovascular risk assessment. Furthermore, despite the demonstrated incremental value of the TyG-WWI over traditional TyG-related parameters in predicting cardiovascular mortality, the TyG-WWI is not intended to serve as a direct alternative to these comprehensive, multifactorial risk models, nor should it be used in isolation during clinical decision-making. Instead, the TyG-WWI should function as a supplementary metric that complements these well-validated tools while always being integrated with clinicians' professional judgment. Incorporating such validated risk assessment tools into a comprehensive decision-making framework guided by clinical expertise helps identify the most appropriate management strategies and promotes effective and safe clinical decisions.

Strengths and limitations

This study has several strengths. First, the large sample size from the NHANES ensures sufficient statistical power and enables more precise estimation of the associations. Second, the use of weighted Cox proportional hazards models and adjustment for multiple confounding factors strengthens the validity of the results. Third, additional analyses, including RCS analysis, subgroup analysis, and sensitivity analysis, provide a more comprehensive understanding of the relationships and enhance the robustness of the findings.

However, several limitations of this study should be acknowledged. First, as an observational study based on NHANES data, a causal relationship between the TyG-WWI and cardiovascular mortality could not be established. Confirmation of causality would require interventional studies to verify whether optimizing

the TyG-WWI through clinical interventions (such as lipid-lowering or glucose-control therapies) can truly reduce the risk of cardiovascular death. Second, a substantial proportion of participants were excluded because of missing data, particularly regarding key variables required for calculating the TyG-WWI. Although most baseline characteristics between the included and excluded adult participants were comparable, notable differences were observed in drinking status and hyperlipidemia, which may introduce selection bias and potentially limit the generalizability of the findings. Third, residual confounding cannot be completely ruled out. Despite rigorous adjustment for key confounding factors (e.g., diabetes, hypertension, smoking), the influence of unmeasured variables—such as detailed dietary patterns, physical activity intensity, or genetic factors—on the results remains a possibility. Additionally, while we calculated the E value to assess the impact of unmeasured confounders, the potential for residual confounding from such factors cannot be entirely eliminated. Fourth, although we excluded early deaths with follow-up ≤ 2 years in sensitivity analyses and observed stable effect sizes for the association between the TyG-WWI and cardiovascular mortality, residual reverse causality cannot be fully eliminated. For example, unrecognized subclinical cardiovascular disease at baseline may simultaneously affect TyG-WWI levels and subsequent mortality risk, thereby confounding the observed association. Future studies should incorporate repeated measurements of the TyG-WWI to more clearly elucidate the temporal relationship between the TyG-WWI and cardiovascular outcomes. Fifth, the inclusion of participants with baseline cardiovascular disease (CVD) may have skewed the cohort's risk profile toward a population with poorer prognosis. However, sensitivity analyses confirmed that the TyG-WWI retained robust predictive ability in individuals without baseline CVD, mitigating concerns about the generalizability of the findings to primary prevention settings. Importantly, this inclusion strategy aligns with clinical reality: reliable risk assessment tools are needed for both primary and secondary prevention populations, and thus, the results hold practical relevance across the spectrum of cardiovascular risk management. Sixth, the study population consists exclusively of U.S. adults. Given that different ethnic groups exhibit variations in insulin resistance patterns, anthropometric distributions, and cardiovascular risk profiles—coupled with potential differences in dietary or genetic backgrounds between Western and non-Western populations—the generalizability of the predictive value of the TyG-WWI to other ethnicities, geographic regions, or healthcare systems remains to be further validated. Finally, the biological mechanisms underlying the predictive ability of the TyG-WWI—particularly how its

composite components interact to reflect cardiovascular risk—require further exploration to enhance its clinical interpretability.

Conclusion

In this large-scale, nationally representative cohort study, for the first time, we confirmed a significant positive linear correlation between the TyG-WWI and the risk of cardiovascular mortality in the general population. Importantly, the TyG-WWI demonstrates greater predictive value than traditional TyG-related indicators do, with improved risk stratification capability and enhanced net reclassification performance (correctly reclassifying 9.1 to 10.4% more participants than traditional indicators do, thereby reducing classification errors). These findings support the use of the TyG-WWI as a supplementary tool for cardiovascular risk screening, although it should be combined with comprehensive patient assessments and other risk factors in clinical practice.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12933-025-02873-8>.

Supplementary Material 1.

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Author contributions

JP-L conceptualized the study, developed the methodology, supervised the project, and administered its execution. JJ-Q wrote the original draft of the manuscript. JJ-Q, JE-L, SX, JQ-Y, HX-Z, YY-Z, and SQ-Y contributed to reviewing and editing the manuscript. JJ-Q and JE-L developed the software. JJ-Q, LX-F, JD-H, and JY-F performed formal analysis and validation. JJ-Q, HT-Z, YJ-Z, and J-Z managed data curation. All authors read and approved the final manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹Department of Endocrinology and Metabolism of the Second Affiliated Hospital, Jiangxi Medical College, Nanchang University, No. 1, Minde Road, Nanchang 330006, Jiangxi, China

²Institute for the Study of Endocrinology and Metabolism in Jiangxi Province, Nanchang 330006, Jiangxi, China

³Department of Cardiovascular Medicine, Jiangxi Provincial People's Hospital, The First Affiliated Hospital of Nanchang Medical College, Nanchang, China

⁴Branch of National Clinical Research Center for Metabolic Diseases, Nanchang 330006, Jiangxi, China

⁵Jiangxi Key Laboratory of Molecular Medicine, The Second Affiliated Hospital of Nanchang University, Nanchang 330006, Jiangxi, China

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